

Task 3: REST - Get Country Based on Country Code

Scenario:

This task involves creating a RESTful web service using Spring Boot that returns a specific country's details when the country code is passed in the URL. The country data is fetched from the XML configuration file (country.xml). The matching is done case-insensitively using the `@PathVariable` annotation. This eliminates the need for a separate service layer.

Controller Code (CountryController.java):

```
package com.cognizant.spring_learn.controller;

import com.cognizant.spring_learn.Country;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
public class CountryController {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);

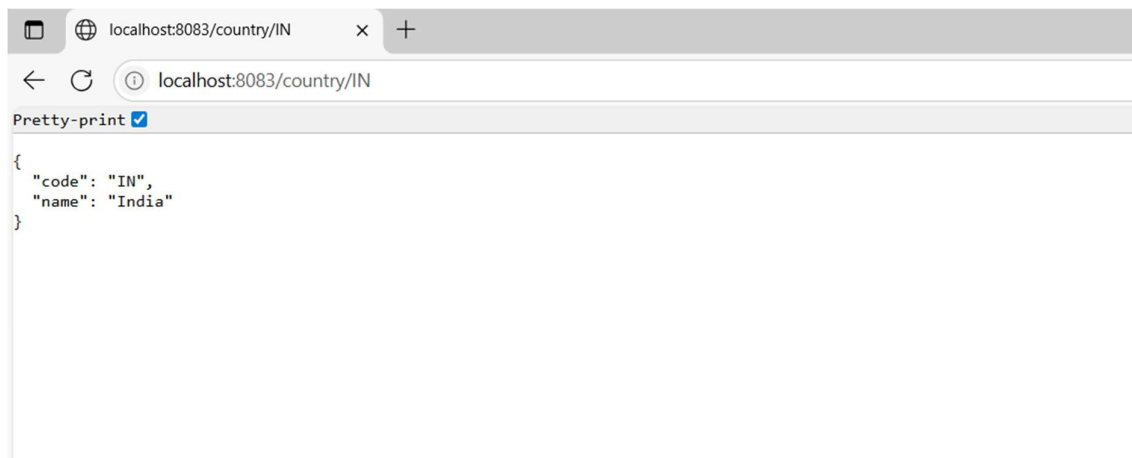
    @GetMapping("/country/{code}")
    public Country getCountry(@PathVariable
        String code) {
        LOGGER.info("START - getCountry with
            code: {}", code);
```

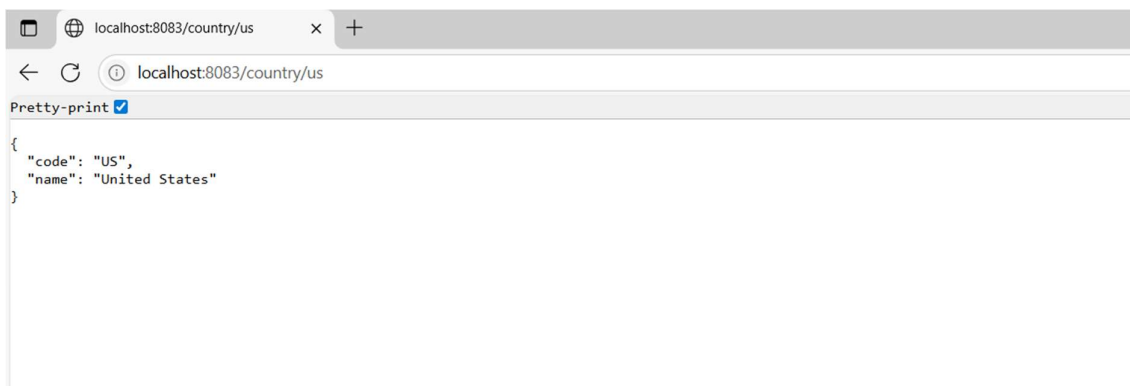
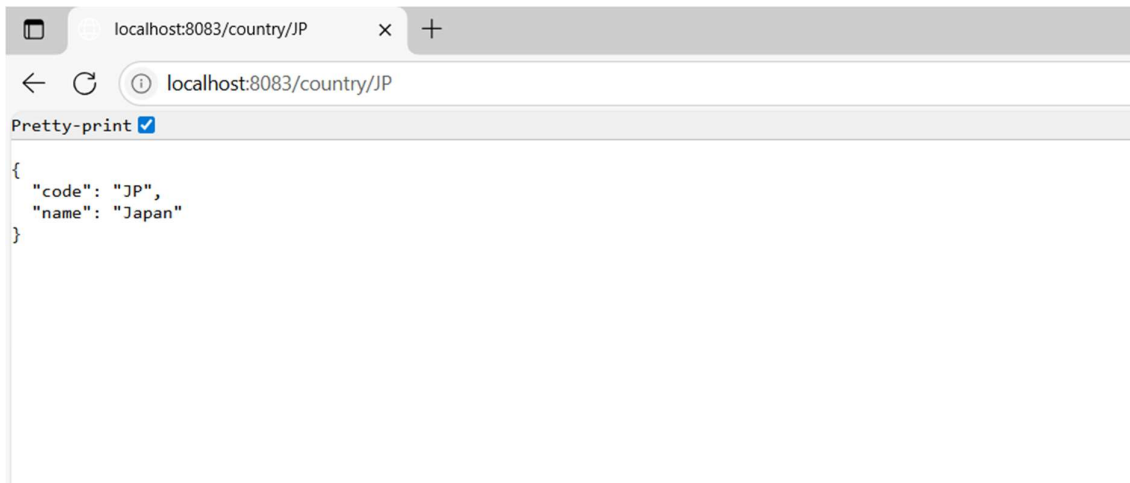
```
        ApplicationContext context = new
ClassPathXmlApplicationContext("country.xml");
        List<Country> countryList =
(List<Country>) context.getBean("countryList");

        Country result = countryList.stream()
                .filter(country ->
country.getCode().equalsIgnoreCase(code))
                .findFirst()
                .orElse(null);

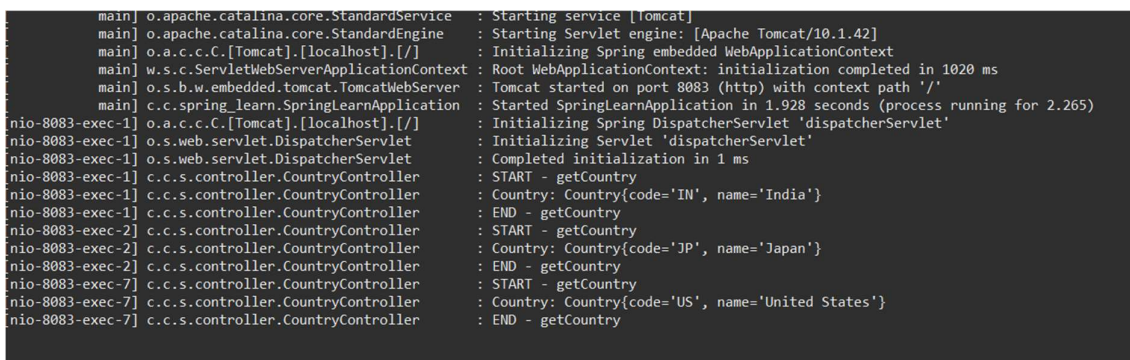
        LOGGER.info("Country Found: {}",
result);
        LOGGER.info("END - getCountry");

        return result;
    }
}
```





Screenshot: Console Logs



Complete Screenshot :



```
:: Spring Boot :: (v3.5.3)

2025-07-11T22:51:24.338+05:30 INFO 20072 --- [main] c.c.spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 21.0.7 with PID 20072 (C:\Users\chait\workspace\spring-learn\spring-learn\src\main\resources)
2025-07-11T22:51:24.344+05:30 INFO 20072 --- [main] c.c.spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2025-07-11T22:51:25.358+05:30 INFO 20072 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2025-07-11T22:51:25.373+05:30 INFO 20072 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2025-07-11T22:51:25.374+05:30 INFO 20072 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.42]
2025-07-11T22:51:25.441+05:30 INFO 20072 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2025-07-11T22:51:25.445+05:30 INFO 20072 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1020 ms
2025-07-11T22:51:25.841+05:30 INFO 20072 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2025-07-11T22:51:25.853+05:30 INFO 20072 --- [main] c.c.spring_learn.SpringLearnApplication : Started SpringLearnApplication in 1.928 seconds (process running for 2.265)
2025-07-11T22:51:52.201+05:30 INFO 20072 --- [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2025-07-11T22:51:52.282+05:30 INFO 20072 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2025-07-11T22:51:52.283+05:30 INFO 20072 --- [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2025-07-11T22:51:52.319+05:30 INFO 20072 --- [nio-8083-exec-1] c.c.s.controller.CountryController : START - getCountry
2025-07-11T22:51:52.426+05:30 INFO 20072 --- [nio-8083-exec-1] c.c.s.controller.CountryController : Country: Country{code='IN', name='India'}
2025-07-11T22:51:52.427+05:30 INFO 20072 --- [nio-8083-exec-1] c.c.s.controller.CountryController : END - getCountry
2025-07-11T22:52:08.210+05:30 INFO 20072 --- [nio-8083-exec-2] c.c.s.controller.CountryController : START - getCountry
2025-07-11T22:52:08.269+05:30 INFO 20072 --- [nio-8083-exec-2] c.c.s.controller.CountryController : Country: Country{code='JP', name='Japan'}
2025-07-11T22:52:08.271+05:30 INFO 20072 --- [nio-8083-exec-2] c.c.s.controller.CountryController : END - getCountry
2025-07-11T22:52:12.893+05:30 INFO 20072 --- [nio-8083-exec-7] c.c.s.controller.CountryController : START - getCountry
2025-07-11T22:52:13.084+05:30 INFO 20072 --- [nio-8083-exec-7] c.c.s.controller.CountryController : Country: Country{code='US', name='United States'}
2025-07-11T22:52:13.084+05:30 INFO 20072 --- [nio-8083-exec-7] c.c.s.controller.CountryController : END - getCountry
```